Afsana Afrin

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Work Authorization: F-1 Visa (Eligible for CPT/OPT; No employer sponsorship required) | Willing to relocate

Summary

PhD student in Computing seeking software engineering roles. Builds machine learning systems, secure full-stack web applications, and automated cloud delivery pipelines in Python, C++, and TypeScript. Seven peer-reviewed publications (three first-author); former lecturer in programming, Data Structures and Algorithms, and Database Design.

Education

Boise State University August 2025 – May 2028 (Expected)
Doctor of Philosophy, Computing (Cybersecurity Emphasis) | Grade Point Average: 4.00/4.00 *Boise, Idaho*

Rajshahi University of Engineering & Technology April 2024
Master of Science, Computer Science & Engineering | Grade Point Average: 3.92/4.00 *Rajshahi, Bangladesh*

Rajshahi University of Engineering & Technology August 2019
Bachelor of Science, Computer Science & Engineering | Grade Point Average: 3.83/4.00 (Rank: 5th of 112) *Rajshahi, Bangladesh*

Theses: Master's — Hyperspectral Image Compression with Deep Neural Networks; Bachelor's — Privacy-Preserving Data Mining

Technical Skills

Languages & Core Competencies: Python, C++, Java, TypeScript, C, SQL, Bash | Data Structures and Algorithms, Problem-Solving

Machine Learning: PyTorch, TensorFlow/Keras, scikit-learn | Deep Learning, Convolutional Neural Networks, Autoencoders, Random Forest, Support Vector Machines, Decision Trees

Software Engineering, Databases & Systems: Git/GitHub, Pull Request Workflows, Property-Based Testing, Code Review, YAML | PostgreSQL, MySQL, SQLite | Linux, Shell Scripting, Networking

Cloud, Containers & Continuous Delivery: Amazon Web Services (AWS: EC2, VPC, IAM, S3), Terraform, Ansible | Docker, Kubernetes, Helm, kind | GitHub Actions Pipelines

Security: Application Security, Threat Modeling, Trivy, Gitleaks, Intrusion Detection, Industrial Control System Security, Digital Forensics

Selected Research [Google Scholar]

- **Physics-Witnessed Intrusion Detection for Industrial Control Systems** — *Manuscript in preparation, 2026*
  - Random Forest plus temporal-logic alarms; eighty percent attack detection on water-treatment data. *Tools:* Python, scikit-learn
- **Resilience-Induced Security Blindness in Industrial Control Systems** — *Manuscript in preparation, 2026*
  - Physical water-distribution testbed showing how fault-tolerant control hides sensor attacks. *Tools:* Python
- **Efficient Hyperspectral Data Compression using 3D Convolutional Autoencoder**
  ***A. Afrin**, M. A. Mamun. 3rd International Conference on Advancement in Electrical and Electronic Engineering, 2024* [Link]
  - Three-dimensional convolutional autoencoder; compression ratio up to 982, about three times the baseline. *Tools:* Python, Keras
- **Enhancing Hyperspectral Image Compression through Stacked Autoencoder Approach**
  ***A. Afrin**, M. R. Haque, M. A. Mamun. 6th International Conference on Electrical Engineering and Information & Communication Technology, 2024* [Link]
  - Stacked autoencoder raising Peak Signal-to-Noise Ratio to 70.43 decibels (baseline 54.46). *Tools:* Python, Keras
- **Privacy Preserving Data Mining Using Non-Negative Matrix Factorization and Singular Value Decomposition**
  ***A. Afrin**, M. K. Paul, A. H. M. S. Sattar. 4th International Conference on Electrical Information and Communication Technology, 2019* [Link]
  - Matrix-factorization data perturbation keeping 95–97 percent classification accuracy. *Tools:* Python

Selected Projects

Application Security & Digital Forensics | *TypeScript, Next.js, PostgreSQL, fast-check, Autopsy* 2025–2026
- Fixed injection, access-control, password-storage, and Cross-Site Request Forgery flaws in a full-stack web application
- Investigated disk, network, mobile, and Internet of Things evidence with hash-verified custody; exposed deliberate evidence wiping

DevOps & Cloud Infrastructure Engineering | *Docker, Kubernetes, GitHub Actions, Terraform, Ansible* 2026
- Shipped a containerized Python service to Kubernetes via a GitHub Actions test, build, and security-scan pipeline
- Provisioned Amazon Web Services infrastructure with Terraform, least-privilege access policies, and Ansible host configuration

Software Development & System Design | *Laravel, PHP, MySQL* | [GitHub 1], [GitHub 2] 2017–2018
- Designed and built two web applications with model–view–controller architecture, relational databases, and role-based access

Professional Experience

Graduate Assistant, Boise State University August 2025 – Present
Privacy and Industrial Control Systems Security Lab, Department of Computer Science *Boise, Idaho*
- Teaching Assistant for Computer Security and Introductory Computer Science; researching industrial control systems security

Lecturer / Assistant Professor, Rajshahi University of Engineering & Technology December 2021 – July 2025
Institute of Information and Communication Technology *Rajshahi, Bangladesh*
- Taught Computer Programming, Data Structures and Algorithms, and Database Design and Management; currently on study leave

Advanced Coursework

Software Engineering, Database Systems, Operating Systems, Computer Networks, Artificial Intelligence, Machine Learning, Data Mining, Software Security, Network Security, Digital Signal & Image Processing, Introduction to Cybersecurity, Digital Forensics